

Sahil Shah

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SUMMARY

Mechanical engineering graduate with hands-on experience in CAD design, technical documentation, and prototype development. Experience in interpreting engineering drawings and translating customer requirements into feasible design solutions. Eager to contribute to cross-functional application engineering teams supporting product design and integration.

EDUCATION

B.S. Mechanical Engineering | University of North Carolina at Charlotte | May 2026

TECHNICAL SKILLS

CAD & Modeling: SolidWorks, Creo Parametric, AutoCAD, Revit

Simulation & Analysis: MATLAB, MathCAD, Blade Element Momentum (BEM) analysis, aerodynamic simulation, QBlade, Creo Simulate FEA

Programming: Python, C++

Fabrication & Prototyping: FDM & SLA 3D printing, rapid prototyping

Software & Tools: Microsoft 365 (Word, Excel, PowerPoint), Prusa Slicer, Arduino IDE

WORK EXPERIENCE

Engineering Intern | M3 Engineering | May 2024 – Aug 2024 |

- Developed CAD layouts for slurry storage tanks and industrial infrastructure systems, producing technical drawings reviewed by senior engineers for feasibility and compliance with project specifications.
- Prepared and organized technical documentation packages including engineering drawings, manuals, and project records
- Collaborated with engineering staff to ensure documentation aligned with design intent and customer deliverable requirements.

Teaching Assistant | **Thermodynamics II** | UNC Charlotte | Aug 2025 – May 2026 |

- Supported 80+ students in understanding complex thermodynamic principles through organized review sessions, study guides, and occasional direct technical instruction.
- Evaluated design exercises and homework to provide constructive feedback.

ENGINEERING PROJECTS

Collegiate Wind Competition 2026

- Designed, fabricated, and tested small-scale wind turbine blades for power production, applying aerodynamic simulation and blade element momentum (BEM) analysis to optimize turbine efficiency and power output.
- Designed and implemented a variable-density internal blade structure. Increased shell thickness and infill near the root to resist bending moments, with a hollow tip section to minimize rotational inertia and centrifugal loading.
- Collaborated within a multidisciplinary team to develop and validate prototype mechanical and electrical systems, contributing to cross-functional project planning and testing coordination.
- Performed iterative design improvements based on test data and simulation results.
- Engineered a vehicle-based test methodology to approximate wind tunnel conditions in the absence of a wind tunnel.

Robot Design Lab | 3-DOF Pick-and-Place System Gripper Design

- Designed and fabricated a custom servo-driven gripper for a 3-DOF robotic pick-and-place system, iterating through three design generations.
- Produced dimensioned CAD drawings and a full component bill of materials, ensuring design feasibility and part compatibility prior to fabrication.
- Validated system performance against functional requirements, applying feasibility analysis and design iteration throughout the development cycle.